

Are major research gaps still present in agentic LLM–time series frameworks for privacy-preserving prediction of drug-induced acute kidney injury?

Yes, major research gaps still persist in privacy-preserving agentic DI-AKI prediction frameworks.

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Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on unresolved gaps in agentic DI-AKI prediction

The meter should read strongly yes. General AKI prediction is well developed, but the combined target framework remains under-evidenced because DI-AKI etiology is hard to label, LLM-agent systems are early, and privacy-preserving ADR prediction with federated LLMs has only limited documented use cases (Tomašev et al., 2019; Yousif & Awdishu, 2022; Guo & Choo, 2024).

Evidence Coverage

Subtopic	Etiology Labels	Temporal Modeling	External Validation	Privacy Evaluation
General AKI ML	3	10	8	GAP
Drug-induced AKI ML	6	4	3	GAP
LLM or agentic AKI	1	2	2	GAP
Synthetic-data DI-AKI	1	1	1	1
Federated LLM for ADR	GAP	GAP	1	1

FIGURE 2 Evidence coverage across agentic privacy-preserving DI-AKI subtopics

The sharpest gap is the **near-empty overlap** between four needs at once: drug-specific causality, longitudinal prediction, LLM-agent reasoning, and formal privacy evaluation. Reviews of DI-AKI state that most AKI models do not ascertain etiology and need clinically adjudicated cases, while federated LLM work for adverse drug reactions remains early-stage with limited real-world examples (Yousif & Awdishu, 2022; Guo & Choo, 2024). Synthetic-data work is even thinner in this exact use case: one vancomycin/ceftazidime-avibactam study found better 30-day prediction after synthetic augmentation, but it also said further validation is needed (Ramazani et al., 2025).

Main Gaps

- **Causal labeling** remains the central gap because DI-AKI is heterogeneous by drug and mechanism, so temporal association alone misclassifies events (Yousif & Awdishu, 2022).
- **Dynamic drug exposure modeling** is still incomplete, since many nephrotoxicity studies remain predose or static and miss postdose temporal risk evolution (Ihsan et al., 2026).
- **Agentic LLM reliability** is early, because current AKI LLM systems still face hallucination risk and often do not output calibrated probabilities (Zhu et al., 2025).
- **Privacy-preserving deployment** lacks direct evidence, as federated LLM-ADR prediction has few documented use cases and focuses on implementation challenges rather than mature validation (Guo & Choo, 2024).

What Exists

General **time-series AKI prediction** is already strong. Continuous and multicenter models can predict AKI 14-48 hours early, often with good external performance, including drug-exposed cohorts and real-time updating frameworks (Tomašev et al., 2019; Alfieri et al., 2023; Heo et al., 2023).

Drug-aware prediction also exists, but mostly in narrow slices. Studies model nephrotoxic-drug cohorts, vancomycin AKI, drug combinations, and medication-informed ICU prediction, yet these systems are usually not agentic, not privacy-preserving, and not built around adjudicated DI-AKI causality (Heo et al., 2023; Aghamirzaei et al., 2025; Wang et al., 2020).

LLM integration appears promising for **interpretability** and external robustness. Multi-agent AKI-Detector outperformed baseline ML on MIMIC-IV, and postoperative AKI LLM models produced human-readable explanations with better external accuracy than comparator ML models (Shi et al., 2024; Zhu et al., 2025).

Bottom Line

Yes, major research gaps are still present in agentic LLM–time series frameworks for privacy-preserving prediction of drug-induced acute kidney injury. The field has strong pieces for AKI forecasting, but it still lacks integrated studies that jointly validate **drug-specific causality**, **longitudinal realism**, **LLM-agent reliability**, and **privacy protection** (Rehman et al., 2025; Ihsan et al., 2026).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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